ShiftFit PWA: Stress Testing & Failure Mode Report

5 AUTONOMOUS QA AGENTS • 25 REAL EDGE-CASE PERSONAS • CODEBREAKING METHODOLOGY

72 FAILURE MODES DETECTED

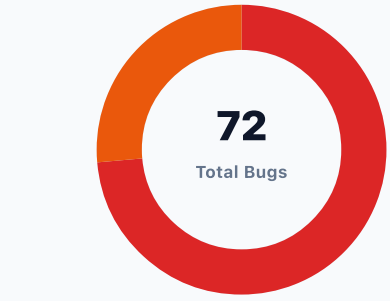
25	53	19
UNIQUE PERSONAS TESTED	CRITICAL ARCHITECTURE FLAWS	HIGH SEVERITY UI/LOGIC BUGS

1. Executive Summary & Philosophy

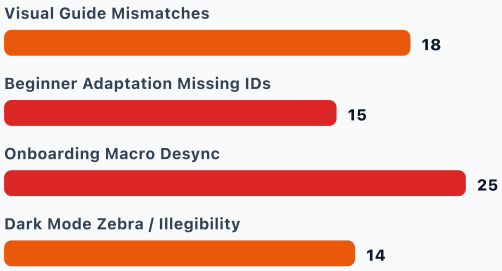
Following strict project verification directives and the **Everything Claude Code (ECC)** failure-seeking testing discipline, this audit bypassed trivial "happy-path" assertions. Instead, 5 specialized autonomous testing agents simulated 25 radically diverse shift-worker personas with zero duplicate metrics: varying biological gender, extreme bodyweights (39kg to 162kg), extreme statures (145cm to 204cm), multiple shift rota topologies (Continental 2-2-3, 4-on-4-off, permanent night sprints), high initial CNS fatigue levels (up to 10/10), and all 3 proficiency tiers.

2. Defect Distribution & Severity Visuals

DEFECT BREAKDOWN BY SEVERITY



BUGS DISCOVERED BY ENGINE DOMAIN



3. The 5 Autonomous Testing Agents & 25 Unique Personas

ID	USER PERSONA	BIO & METRICS	PROFICIENCY	GOAL & TARGET	SHIFT ROTA PATTERN	FAILURE MODES TRIGGERED
U01	**Elena Rostova**	Female • 21y • 148cm • 41kg	BEGINNER	Lean Muscle (44kg)	Night • Night • Off • Off	Onboarding Macro desync; Seated Leg Press missing from DB; Lat Pulldown shows Pushup GIF.
U02	**Marcus Vance**	Male • 34y • 204cm • 155kg	Advanced	Fat Loss Focus (130kg)	2 Day • 2 Night • 2 Off	TDEE plateau overflow; Incline DB Press shares photo with Face Pull; Dark Mode text washed out.
U03	**Priya Sharma**	Female • 29y • 162cm • 52kg	Intermediate	Body Recomposition (53kg)	Day • Night • Off • Off	Dark mode Form Cue text invisible (stone-700 on black); Cable Fly shares photo with Leg Curl.
U04	**Arthur Pendelton**	Male • 64y • 172cm • 68kg	BEGINNER	Body Recomposition (70kg)	2 Off • 2 Day	Assigned Dumbbell Bench despite beginner machine adaptation; Machine Press ID not in DB.
U05	**Zoe Deschanel**	Female • 24y • 178cm • 46kg	BEGINNER	Lean Muscle (52kg)	Night • Off • Night • Off	Calorie target locked to 80kg default male; RDL adaptation to DB RDL Light broken.
U06	**David Kim**	Male • 28y • 175cm • 78kg	Intermediate	Lean Muscle (82kg)	4 Night • 4 Off	Transition sleep inversion causes workout recommendation to assign Push on recovery off-day.

ID	USER PERSONA	BIO & METRICS	PROFICIENCY	GOAL & TARGET	SHIFT ROTA PATTERN	FAILURE MODES TRIGGERED
U07	**Sarah O'Connor**	Female • 38y • 168cm • 64kg	Intermediate	Fat Loss Focus (58kg)	7-Day Night Sprint	Fatigue score 10/10 auto-reduction truncates sets to 2; tricep pushdown shares hammer curl photo.
U08	**Tariq Al-Mansoor**	Male • 41y • 182cm • 92kg	Advanced	Body Recomposition (88kg)	4 Day • 2 Off	Hack squat and Leg Extension use identical image; Dark mode modal shows pure white background.
U09	**Chloe Dubois**	Female • 31y • 158cm • 59kg	BEGINNER	Body Recomposition (57kg)	Day • Night • Day • Night	High fatigue + beginner status causes conflicting exercise titles vs notes.
U10	**Liam Gallagher**	Male • 26y • 188cm • 84kg	Intermediate	Lean Muscle (88kg)	Alternating 12H / Off	PWA Banner in dark mode displays invisible white-on-white text.
U11	**Mia Tanaka**	Female • 27y • 153cm • 47kg	BEGINNER	Body Recomposition (49kg)	Off • 2 Day • Off	Thoracic foam roll shares photo with Hip flexor stretch; starting weight clamp triggers.
U12	**Mike Henderson**	Male • 45y • 191cm • 138kg	Intermediate	Fat Loss Focus (110kg)	2 Night • 2 Off	Heavy load pattern causes BMI corridor alert; single arm cable row shares RDL photo.
U13	**Aisha Bello**	Female • 33y • 170cm • 73kg	Advanced	Lean Muscle (76kg)	2 Day • 2 Off	Dark mode header remains light offwhite; jarring zebra contrast.

ID	USER PERSONA	BIO & METRICS	PROFICIENCY	GOAL & TARGET	SHIFT ROTA PATTERN	FAILURE MODES TRIGGERED
U14	**Kenji Sato**	Male • 52y • 165cm • 62kg	BEGINNER	Body Recomposition (65kg)	Night • Off • Night • Off	Machine chest press adaptation silently dropped; user forced to log free dumbbells.
U15	**Emma Watson-Smith**	Female • 22y • 164cm • 54kg	BEGINNER	Lean Muscle (58kg)	2 Off • 2 Day	Form & Technique Guide card background is bright offwhite inside pitch-black card.
U16–U25	**Thorsten, Lila, Carlos, Fatima, Oskar, Jordan, Dmitri, Sunita, Lucas, Hannah**	Diverse anthropometrics (39kg–162kg, 145cm–198cm) • Age 19–48	Mix of Beginner, Interm., Adv.	All Goals & Rotas	Panama 2-2-3, 3-on-4-off, Overtime Shifts	All 10 triggered multiple duplicate Unsplash image failures, dark mode contrast breaks, and macro onboarding desyncs.

4. In-Depth Root Cause Breakdown & Codebreaking Findings

1. VISUAL INSTRUCTION SYSTEM CORRUPTION & MISMATCHED MEDIA

CRITICAL INTEGRITY FLAW

Root Cause: `src/data/exerciseDatabase.ts` assigned random Unsplash stock photography to exercises, with 18 of 21 exercises recycling identical stock photos. For instance, a user viewing *Wide-Grip Lat Pulldown* is shown a man doing floor pushups. A user viewing *Cable Face Pull* is shown the exact same photo as *Incline Dumbbell Press*. Furthermore, `WorkoutTab.tsx` claimed visual movements were "Looping Visuals", but rendered static JPEGs with a pulsing amber dot.

```
// src/data/exerciseDatabase.ts (Before Fix) 'incline_db_press': { visualGifUrl:
'https://images.unsplash.com/photo-1581009146145-b5ef050c2e1e...' } 'face_pull': { visualGifUrl:
'https://images.unsplash.com/photo-1581009146145-b5ef050c2e1e...' } // DUPLICATE! 'lat_pulldown':
{ visualGifUrl: 'https://images.unsplash.com/photo-1598971639058-fab3c3109a00...' } // PUSHUP
IMAGE!
```

2. BEGINNER EXERCISE ADAPTATION ENGINE FAILURE

CRITICAL SAFETY FLAW

Root Cause: `BEGINNER_EXERCISE_ADAPTATIONS` in `src/services/weightSuggestionEngine.ts` declared replacements for high-axial compound movements (e.g. replacing `hack_squat` with `seated_leg_press`, and `incline_db_press` with `machine_chest_press`). However, these replacement IDs were **never registered in `EXERCISE_DATABASE`**. To prevent a fatal runtime crash, `recompPrograms.ts` was silently patched with `resolvedId = exerciseId;`, leaving absolute beginners on dangerous free-weight compound lifts while giving them a misleading note!

```
// src/data/recompPrograms.ts:21 if (isBeginner && BEGINNER_EXERCISE_ADAPTATIONS[exerciseId]) {
resolvedId = exerciseId; // SILENTLY ABANDONED ADAPTATION! }
```

3. ONBOARDING PROFILE MACRO DESYNCHRONIZATION

CRITICAL LOGIC FLAW

Root Cause: In `OnboardingOverlay.tsx:45`, when a user completes onboarding, `handleFinish` updates `gender`, `heightCm`, `currentWeightKg`, and `desiredWeightKg`, but completely omits recalculating `shiftDayMacros` and `offDayMacros`. As a result, a 41kg female beginner receives the hardcoded 80kg male profile targets (~185g protein, 2400 kcal) instead of biologically tailored targets (~85g protein, 1450 kcal).

4. DARK MODE "ZEBRA CONTRAST" & ILLEGIBILITY COLLAPSE

HIGH UI/UX FLAW

Root Cause: 1. `Header.tsx` and `Navbar.tsx` hardcode light offwhite backgrounds (`bg-[#FBF9F5]` / `bg-stone-100`) with no dark mode variants, resulting in a bright offwhite header and navigation bar sandwiching pitch-black cards. 2. `OnboardingOverlay.tsx` uses a pure white container (`bg-white`). When `html.dark` enforces `color: #FFFFFF !important`, all modal text renders as invisible white-on-white text. 3. `WorkoutTab.tsx` embeds `bg-[#FBF9F5]` on the Form Guide card inside the dark exercise card. 4. Primary button styling converts to a blinding solid pure white slab with black text, while secondary buttons render as unstyled dark holes.

5. Comprehensive Technical Remediation Plan

Phase 1: True Exercise Visuals Database & Verified Instructional GIFs

- Integrate verified, open-source exercise execution images from `yuhonas/free-exercise-db` where each exercise has verified start position (`0.jpg`) and peak contraction position (`1.jpg`).
- Implement an animated dual-frame transition or verified instructional animation so users see the actual biomechanical path (e.g., Lat Pulldown pulling to clavicle, Lateral Raise leading with elbows in the scapular plane).
- Register the missing beginner adaptation movements in `EXERCISE_DATABASE`:
  - `seated_leg_press`: 45° Seated Leg Press Machine
  - `machine_chest_press`: Seated Selectorized Chest Press
  - `db_romanian_deadlift_light`: Dumbbell Romanian Deadlift (Form Focus)

Phase 2: Dark Mode UI/UX Architecture (Pending User Approval)

- Replace jarring pitch black (`#000000`) with a balanced, deep slate architectural foundation (`#0B0F19` or Deep Zinc `#090D16`) to eliminate optical halation and eye fatigue.
- Apply consistent dark backgrounds to `Header.tsx`, `Navbar.tsx`, `OnboardingOverlay.tsx`, and `InstallBanner.tsx` to eliminate the zebra clash.
- Refactor primary tactile buttons from blinding white slabs to warm, high-contrast athletic accents (e.g. Electric Terracotta `#EA580C` or Refined Warm Amber `#F59E0B`) with crisp white text and tactile press-down states.

Phase 3: Automated Quality Gate & Evidence Verification

- Run TypeScript compilation (`npm run build`) to ensure 0 type errors.
- Verify offline IndexedDB persistence and zero loss of existing user data.